Imperial College London
BSc/MSci EXAMINATION June 2025

This paper is also taken for the relevant Examination for the Associateship

PRINCIPLES OF INSTRUMENTATION

For 3rd/4th-Year Physics Students

Wednesday, 28th May 2025: 10:00 to 12:00

The paper consists of **three** *questions – please answer* **all three***.*
A table of Laplace transforms is included at the end of the paper.

Marks shown on this paper are indicative of those the Examiners anticipate assigning.

General Instructions

Complete the front cover of each of the 3 answer books provided.

If an electronic calculator is used, write its serial number at the top of the front cover of each answer book.

USE ONE ANSWER BOOK FOR EACH QUESTION.

Enter the number of each question attempted in the box on the front cover of its corresponding answer book.

Hand in 3 answer books even if they have not all been used.

You are reminded that Examiners attach great importance to legibility, accuracy and clarity of expression.

1. A particle detector contains a liquid scintillator which emits optical photons in response to particle interactions. A fraction of this light is collected by optical fibres immersed in the liquid, with photons detected several metres away by silicon photomultipliers (SiPMs).

(i) (a) List five factors determining the energy threshold of the detector, i.e. the minimum energy deposited in the liquid that allows the particle interaction to be detected.

ANSWER: [BOOKWORK]

Give [1] each for answers in this list or others that apply (maximum 5 marks):

- Scintillation yield or similar; scintillation wavelength; bulk absorption
- Number density of fibres; length of fibres within liquid
- Optical coupling to fibre, refractive index mismatch (either end); attenuation
- SiPM coupling/alignment to fibre; SiPM photon detection efficiency; SiPM dark rate; readout noise (frontend or backend).

(b) Explain briefly how a SiPM works, and how its performance compares to that of a photomultiplier tube (PMT) – mention three key parameters.

ANSWER: [BOOKWORK]

SiPMs are solid-state photodetectors consisting of an array of single-photon avalanche diodes (SPADs) operating in Geiger mode. A photon interacting in a SPAD causes an electron avalanche that develops until it is limited by a quenching resistor [2].

Comparison of performance parameters: give [1] each for answers in this list or others that apply (maximum 3 marks):

- A SiPM is characterised by a photon detection efficiency (PDE) rather than just a quantum efficiency (QE) to account for the fill-factor of the microcells. For visible light SiPMs can achieve somewhat higher PDEs than PMTs
- The gains of the two devices are comparable
- SiPM linearity degrades with microcell occupancy, while PMT linearity tends to be related to voltage-divider saturation and space charge effects; at low light levels both are quite linear (dark rate mentioned separately)
- SiPMs are faster than most PMTs
- SiPMs have much larger dark count rates than PMTs
- SiPMs are much physically smaller, require lower supply voltages, are insensitive to magnetic fields, etc.

[10 marks]

(ii) These SiPMs have output resistance of $\sim$300 kΩ, capacitance of 100 pF, and gain $G = 2\times10^6$. Each device is connected to a long transmission line with 50 Ω

[This question continues on the next page . . .]

characteristic impedance and then to an amplifier with $50\,\Omega$ input impedance and voltage gain of 20 dB.

(a) Define characteristic impedance Z_0 and state how to calculate this for a transmission line. Do you expect the recorded data to show reflections in the signal? Justify your answer.
ANSWER: [SEEN]
Characteristic impedance (Z_0) of a transmission line is the ratio of the instantaneous amplitudes of voltage and current of a wave traveling along the line in the absence of reflections. It is determined by the geometry and materials of the transmission line and is independent of its length. For inductance and capacitance per unit length L' and C', we have $Z_0 = \sqrt{L'/C'}$.
We do not expect a reflection: although the sensor output impedance is much larger than that of the cable and the load, the line is matched at the load end so there is no signal coming back to the source.

(b) At the high frequencies supporting our SiPM signals ($\sim$500 MHz) the attenuation of the cable is 0.6 dB/m; what is the reduction in voltage amplitude after 20 m of cable? What causes these losses?
ANSWER: [SIMILAR TO SEEN]
Total attenuation in 20 m of cable at 0.6 dB/m is 12 dB, which corresponds to a voltage attenuation factor of $10^{(12/20)} = 4$. The loss is due to dielectric as well as conductor losses.

(c) Estimate the voltage amplitude of the single photoelectron response at the amplifier output. What practical source of single photoelectron signals allows us to easily measure this experimentally in SiPMs?
ANSWER: [UNSEEN]
The total charge from a single photoelectron is $Q = 2 \times 10^6 \times e = 3.2 \times 10^{-13}$ C;
The voltage amplitude is generated by the sensor capacitance: $V = Q/C = 3.2$ mV (naturally, this makes assumptions about the timing profile of the current pulse and the various SiPM response time constants but we shall ignore these.)
We get $1/4$ of this at the end of the cable and the amplifier has gain of 20 dB ($10\times$), so:

$$V_{out} = 3.2/4 \times 10 = 8 \text{ mV}\,.$$

We can easily measure this from the dark counts since SiPMs have very high dark rates; since they also have excellent resolution this is very easy to measure.

[10 marks]

(iii) A 12-bit analogue-to-digital converter (ADC) is used to measure the amplified SiPM signals with a digital resolution of 1 mV. Determine:

(a) The maximum digitisable input signal in volts.
ANSWER: [SIMILAR TO SEEN]

[This question continues on the next page . . .]

A 12-bit ADC has $2^{12} = 4096$ discrete levels. Given the digital resolution of 0.001 V per level, the maximum signal is $4096 \times 0.001 = 4.096$ V [2]. Accept ± 2.048 V with mention to unipolar input.

(b) The dynamic range for the digitised signal in dB.
ANSWER: [SIMILAR TO SEEN]
The dynamic range is $20 \log_{10} 2^{12} = 20 \log_{10}(4096) = 72.2$ dB [2].

(c) The maximum quantisation error.
ANSWER: [SIMILAR TO SEEN]
The maximum quantisation error for an ADC is half of the least significant bit (LSB), i.e. 1 mV/2 = 0.5 mV [2].

(d) The RMS quantisation noise added.
ANSWER: [SIMILAR TO SEEN]
This is equal to LSB / $\sqrt{(12)}$ = 1 mV/ $\sqrt{(12)}$ = 0.289 mV (rms) [2].

(e) If the electronics noise from the sensor plus amplifier is 20 μV_{rms} (referred to the amplifier input) what is the signal-to-noise ratio for the measurement of single photoelectron signals?
ANSWER: [SIMILAR TO SEEN]
The electronics noise is 0.2 mV_{rms}, which we must add quadratically to the digitisation noise:

$$v_n = \sqrt{0.289^2 + 0.2^2} = 0.35\ mV_{rms}\,.$$

Hence,

$$SNR = \frac{8\ mV}{0.35\ mV} = 22.9\,.$$

[10 marks]

(iv) Fast digitisers operating at 500 MS/s record the signal waveforms.

(a) State the Shannon-Nyquist theorem and indicate the Nyquist frequency of the digitised signal.
ANSWER: [BOOKWORK]
Nyquist theorem: a continuous signal can be completely represented in its sampled form and perfectly reconstructed if it is sampled at a rate that is at least twice the highest frequency present in the signal. The Nyquist frequency is half of the sampling rate of the digitiser, so f_N = 250 MHz.

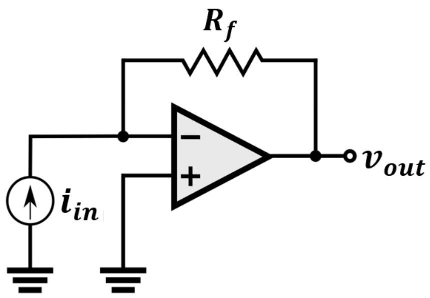
(b) Discuss briefly what filtering might be required for proper sampling of our signals.
ANSWER: [EXTENSION OF SEEN]
Our signals have a frequency spectrum extending considerably above this so we need to use an anti-aliasing filter with cutoff below 250 MHz or use a higher sampling rate. Since the spectrum extends significantly above the Nyquist frequency, a very sharp low-pass filter is required (not a simple RC circuit); examples include Bessel, Chebyshev, Butterworth,...

[10 marks]

[This question continues on the next page . . .]

[Total 40 marks]

2. A transimpedance amplifier (TIA) is often used to convert a low-level current from a photosensor such as a silicon photomultiplier, which generally has high capacitance and high output impedance, into a measurable voltage signal. This is often preferable to converting the current using a simple load resistor R_L. A simplified TIA circuit is shown in the figure below.



(i) Derive the voltage output of the TIA, justifying your assumptions. [5 marks]

ANSWER: [EXTENSION OF SEEN]
Since there is no current into the inverting input, the current flows through the feedback resistor [2]; the inverting input is a virtual ground [1], so

$$v_{out} = -R_f\, i_{in}\,. \quad [2]$$

(ii) Compare the signal-to-noise ratio for a signal current $i_{in}(t)$ achieved with a TIA and with a load resistor R_L. You may assume that both the current source and the op-amp have negligible intrinsic noise. Explain why using a TIA is nonetheless advantageous, referring to both its input and output impedances.
[12 marks]

ANSWER: [UNSEEN]
The Johnson (thermal) noise from a resistor R is $v_n = \sqrt{4k_BTRB}$, where k_B is the Boltzmann constant, T is the temperature and B is the measurement bandwidth [2]. This is what we obtain at the output of the TIA, since the inverting input is a virtual ground: $v_n = \sqrt{4k_BTR_fB}$. This yields:

$$\mathrm{S/N} = \frac{v_{out}}{\sqrt{4k_BTR_fB}} = \frac{i_{in}\sqrt{R_f}}{\sqrt{4k_BTB}}\,. \qquad [2]$$

Essentially the same result applies for the voltage developed over a load R_L:

$$\mathrm{S/N} = \frac{R_L\, i_{in}}{\sqrt{4k_BTR_LB}} = \frac{i_{in}\sqrt{R_L}}{\sqrt{4k_BTB}}\,. \qquad [2]$$

[This question continues on the next page . . .]

However, to achieve high gain from a load resistor we need a high resistance value [1], so we essentially have a circuit with high output impedance which will be easily loaded (and hence deliver reduced gain) once connected to something else [1]. The TIA op-amp offers low output impedance and avoids this problem [1].

Looking now at the input impedance, the TIA has low input impedance as is desirable to measure a current source (since we are connecting the source to a virtual ground) [1], while the load resistor offers a higher impedance – especially if we need high gain [1] – which means we are loading the sensor [1]. (Note that, in the case of a SiPM, the sensor output impedance actually depends on the number of fired micro-cells...)

(iii) Explain why choosing the TIA feedback resistor value is a trade-off between sensitivity and range. How would you determine a reasonable value of R_f knowing that you need to measure currents up to some value i_{max}? Then suggest a trivial modification to the circuit to make it a programmable-gain TIA capable of measuring larger currents. [7 marks]

ANSWER: [UNSEEN]
A large feedback resistor improves sensitivity (cf. S/N calculation above) [1], but the op-amp will clip the signal at its supply (rail) voltages $\pm V_S$ [2]. So, we might choose $R_f = V_S/i_{max}$ [2]. A trivial modification is to add further resistors in parallel to R_F, each with inline with a switch, bringing the equivalent feedback resistance down in defined steps [2].

(iv) From the point of view of bandwidth, what would the advantage be compared with a load resistor given that the photosensor has typically a large capacitance? [6 marks]

ANSWER: [UNSEEN]
Using a load resistor means that the RC time constant of the circuit will be large, since C is large and R_L must also be large to obtain a large signal [2] – and this implies a lower bandwidth [2]. The TIA provides a low output impedance, and the bandwidth of this circuit is determined by the gain-bandwidth product of the amplifier and the feedback resistor [2].

[Total 30 marks]

3. (i) A linear sensor is described by the following differential equation, where $y(t)$ is the forcing function and $x(t)$ is the sensor response:

$$a_1 \frac{\mathrm{d}x}{\mathrm{d}t} + a_0 x = b_0 y \, .$$

(a) Show that the transfer function can be written in the form

$$G(s) = \frac{k_s}{\tau s + 1} \, .$$

Express k_s and τ in terms of the constants a_0, a_1 and b_0.
ANSWER: [SIMILAR TO SEEN]
Taking Laplace transforms [1]

$$a_1 s X(s) + a_0 X(s) = b_0 Y(s)$$

$$X(s)(a_1 s + a_0) = b_0 Y(s)$$

and, by definition of transfer function [1],

$$G(s) = \frac{X(s)}{Y(s)} = \frac{b_0}{a_1 s + a_0} = \frac{b_0/a_0}{a_1/a_0 s + 1} = \frac{k_s}{\tau s + 1} \, ,$$

where we defined $k_s = b_0/a_0$ and $\tau = a_1/a_0$ [2].

(b) Give the common names for the constants k_s and τ and their physical interpretation in terms of the sensor's time-domain behaviour. How is τ related to the frequency response of the sensor?
ANSWER: [BOOKWORK]

- k_s is the *static sensitivity* [1]; it relates the output to the input in the steady-state condition, i.e. for $dx/dt = 0$ in the equation given we see that $x = k_s y$. For a first-order system we expect this output to be reached only asymptotically since $dx/dt \propto (y - x)$, implying that $dx/dt \to 0$ only as $t \to \infty$ [2].
- τ is the *time constant* [1]; it gives the time taken for the output to reach $(1 - 1/e)$ of the steady-state value assuming the input to be a step function. $1/\tau$ gives the angular frequency at which the sensor's frequency response falls to $1/\sqrt{2}$ of the nominal value; the frequency range from 0 to $1/\tau$ defines the bandwidth of the sensor [2].

(c) For a sensor with $a_0 = 10$, $a_1 = 5$ and $b_0 = 100$ (in appropriate units) determine the sensor's response $x(t)$ to a unit-step at the input. How long does the output take to reach 95% of the steady-state value?
ANSWER: [UNSEEN]
For the parameters given we have $k_s = b_0/a_0 = 10$ and $\tau = a_1/a_0 = 0.5$.
For a unit step input we have $y(t) = u(t) \therefore Y(s) = 1/s$.

[This question continues on the next page . . .]

$$X(s) = G(s)Y(s) = \frac{k_s}{s(\tau s + 1)} = \frac{k_s/\tau}{s(s + 1/\tau)} . \quad [2]$$

Using the table transforms given at the end of the paper,

$$x(t) = \mathcal{L}^{-1}\{X(s)\} = \frac{k_s}{\tau}\frac{1}{1/\tau - 0}\left(e^{-0} - e^{-t/\tau}\right) .$$

$$x(t) = k_s\left(1 - e^{-t/\tau}\right) . \quad [2]$$

For $t \to \infty$, $x(t) \to k_s$; we need the time t_{95} taken to reach $0.95k_s$, so

$$0.95k_s = k_s\left(1 - e^{-t_{95}/\tau}\right)$$

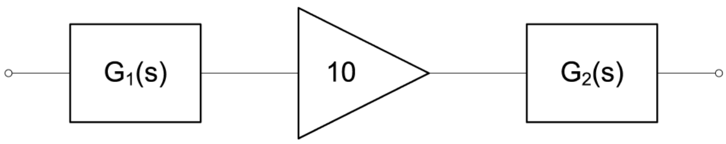
$$e^{-t_{95}/\tau} = 0.05$$

$$t_{95} \approx 3\tau = 1.5 \text{ time units.} \quad [2]$$

[16 marks]

(ii) In the figure below, a first-order sensor is connected to an ideal amplifier of gain 10 and drives a first-order output transducer. The transfer functions for the first-order elements are:

$$G_1(s) = \frac{10^{-3}}{s + 0.5}, \quad G_2(s) = \frac{20}{s + 2} .$$



(a) Write the transfer function for the system. With reference to your answer, does the system behaviour change if the amplifier is moved to the right of the output transducer? Is this physically realistic? Explain your answer.

ANSWER: [SIMILAR TO SEEN]
The amplifier has a transfer function of 10, hence the system function is

$$G(s) = 10G_1(s)G_2(s) = 10\frac{10^{-3}}{s + 0.5}\frac{20}{s + 2} = \frac{0.2}{(s + 0.5)(s + 2)} . \quad [2]$$

Since the above expression is commutative, in principle it does not matter in which order the components lie. However, this is not physically realistic: the amplifier takes a voltage input and produces a voltage output. The output transducer takes a voltage input and produces some physical action or observable, not an electrical signal. Hence we cannot place the amplifier after the transducer [2].

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(b) Determine the system's response to a step function $u(t)$ at the input.
ANSWER: [SIMILAR TO SEEN]
For $y(t) = u(t)$ we have $Y(s) = 1/s$ [1]:

$$X(s) = G(s)Y(s) = \frac{0.2}{s(s+0.5)(s+2)}\,.$$

We need to simplify with partial fractions:

$$X(s) = \frac{A}{s} + \frac{B}{s+0.5} + \frac{C}{s+2}\,. \quad [2]$$

Evaluate $sX(s)|_{s=0}$:

$$\frac{0.2}{(s+0.5)(s+2)} = A + \frac{Bs}{s+0.5} + \frac{Cs}{s+2} \therefore A = 0.2\,. \quad [2]$$

Evaluate $(s+0.5)X(s)|_{s=-0.5}$:

$$\frac{0.2}{(s(s+2)} = \frac{A(s+0.5)}{s} + B + \frac{C(s+0.5)}{s+2}\,,$$

$$\frac{0.2}{-0.75} = B \therefore B = \frac{4}{15}\,. \quad [2]$$

Evaluate $(s+2)X(s)|_{s=-2}$:

$$\frac{0.2}{(s(s+0.5)} = \frac{A(s+2)}{s} + \frac{B(s+2)}{s+0.5} + C\,,$$

$$\frac{0.2}{3} = C \therefore C = \frac{1}{15}\,. \quad [2]$$

Therefore,

$$X(s) = \frac{1}{5s} - \frac{4}{15(s+0.5)} + \frac{1}{15(s+2)}\,.$$

$$x(t) = \mathcal{L}^{-1}X(s) = \frac{u(t)}{5} - \frac{4}{15}e^{-t/2} + \frac{1}{15}e^{-2t}\,. \quad [1]$$

[14 marks]

[Total 30 marks]

$f(t)$	$F(s) = \mathcal{L}\{f(t)\}$
$\delta(t)$	1
$u(t)$	$\frac{1}{s}$
t^n for $n > 0$	$\frac{n!}{s^{(n+1)}}$
e^{-at}	$\frac{1}{s+a}$
te^{-at}	$\frac{1}{(s+a)^2}$
$\frac{1}{b-a}(e^{-at} - e^{-bt})$	$\frac{1}{(s+a)(s+b)}$
$\sin(\omega t)$	$\frac{\omega}{s^2+\omega^2}$
$\cos(\omega t)$	$\frac{s}{s^2+\omega^2}$
$e^{-at}\sin(\omega t)$	$\frac{\omega}{(s+a)^2+\omega^2}$
$e^{-at}\cos(\omega t)$	$\frac{s+a}{(s+a)^2+\omega^2}$

Figure 1: Laplace transforms.